Total Printed Pages: 02

Roll No. :.....

ITC-101

Examination –May- 2024

B.Tech. I/II Sem: Common to all Branches

Python Programming

Time: 2 Hrs

Max. Marks: 60
Min. Marks: 19

Note: Total number of questions are 05. All Questions are compulsory. Each Question has 4 parts (a, b, c, d). Part a & b are compulsory while Part c & d has internal Choice. Assume missing data, if any.

Word limit be observed as follows:

Part a & b – Objective type/Fill in the blanks/One sentence answer.

Part c – Max 100 words and Part d – Max 400 words.

Word limit NOT to be followed for diagram, numerical, derivation.

			Marks	BL	CO
Q.1	(a)	Fill in the blank: In Python, the += operator is used for _____ assignment.	02	1,2	1
	(b)	Define interactive shell and IDLE.	02	1,2	1
	(c)	Explain the difference between mutable and immutable variables in Python. Provide examples to illustrate each.	03	1,2	1
		OR			
		Describe the numerical types available in Python and their respective use cases. Provide examples of each type.	03	1,2,3	1
	(d)	Discuss the importance of indentation and comments in a Python program. How do they enhance code readability and maintainability? Provide examples to illustrate their usage.	05	1,2,3	1
		OR			
		Explain the concept of operators and expressions in Python. Provide examples of arithmetic, comparison, and logical operators along with their use in expressions.	05	1,2,3	1
Q.2	(a)	The elif statement is short for _____ and is used to check for multiple conditions after an initial if statement.	02	1,2	2
	(b)	Define a range in Python.	02	1,2	2
	(c)	Explain the difference between the break, continue, and pass statements in Python. Provide examples to illustrate their usage.	03	1,2,3	2
		OR			
		With the help of an example describe the concept of nested loops in Python.	03	1,2,3	2
	(d)	Write a Python program to find the sum of all numbers between 1 and 100 that are divisible by 3.	05	1,2,3,6	2
		OR			
		Write a Python program to find the factorial of a given number using a for loop.	05	1,2,3,6	2
Q.3	(a)	Fill in the blank: In Python, the _____ operator is used to access individual characters in a string by their index positions.	02	1,2	3
	(b)	Define slicing in Python strings.	02	1,2	3

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	(c)	Explain the process of converting a string to a number and vice versa in Python. Provide example to illustrate this conversion.	03	1,2,3	3
		OR				
		Describe the role of the 'os' and 'sys' modules in Python for manipulating files and directories. Explain with example.	03	1,2,4	3
	(d)	Write a Python program to read numbers from a text file, calculate their sum, and write the result to another file.	05	1,2,4,6	3
		OR				
		Discuss the significance of the CSV (Comma-Separated Values) and tab-separated file formats in Python for handling structured data. Provide examples to demonstrate their usage.	05	1,2,3	3
Q.4	(a)	In Python, a list is a collection of _____ separated by commas and enclosed within square brackets.	02	1,2	4
	(b)	Define a tuple in Python.	02	1,2	4
	(c)	Explain the difference between a list and a tuple in Python. When would you use a list over a tuple, and vice versa?	03	1,2,4	4
		OR				
		Discuss the key characteristics of dictionaries in Python and their significance in storing and retrieving data. Provide examples to illustrate the usage of dictionaries.	03	1,2,3	4
	(d)	Write a Python program to sort a list of numbers in ascending order using the 'sorted()' function.	05	1,2,3, 6	4
		OR				
		Write a Python program to create a dictionary containing student names as keys and their corresponding scores as values. Then, calculate the average score of the students.	05	1,2,3, 4,6	4
Q.5	(a)	Fill in the blank: In object-oriented programming (OOP), a class is a blueprint for creating _____ in Python.	02	1,2	5
	(b)	Define inheritance in the context of object-oriented programming.	02	1,2	5
	(c)	Explain the concepts of overloading and overridingin Python classes. Provide example to illustrate each concept.	03	1,2,4	5
		OR				
		Explain the concept of data hiding in object-oriented programming. How does it contribute to encapsulation? Provide an example to illustrate.	03	1,2, 3	5
	(d)	Write a Python program to create a class called Rectangle with attributes length and width. Include methods to calculate the area and perimeter of the rectangle.	05	1,2,3, 6	5
		OR				
		Discuss the importance of exception handling in Python programming. Explain the purpose of the try, except, and finally clauses in handling exceptions. Provide examples to demonstrate exception handling.	05	1,2,3	5

Bloom's Taxonomy Levels (BL): 1-Remembering, 2- Understanding, 3- Applying, 4- Analysing, 5- Evaluating, 6-Creating

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